BUILDING THE STANDARD MODEL LAGRANGIAN FROM THE HQIV DISCRETE ACTION: A SECTOR-BY-SECTOR LEAN BRIDGE

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ABSTRACT. **Position in the publication chain.** This is the fifth paper in the HQIV preprint series and the second of the Tier-2 Standard-Model layer. Tier 0 fixes the framework [2, 4, 6]; Tier 1 supplies the variational spine [7, 5, 3]; Tier 2 begins with the electroweak and mass ladder [1], continues with the dynamic TUFT/Hopf mass synthesis [8], and the present paper closes the loop: every sector of the textbook Standard-Model Lagrangian $\mathcal{L}_{\mathrm{SM}}$ is identified, sector by sector, with projections of the HQIV *discrete* action already proved upstream. Tier-0 records and the published Tier-1 companions [7, 5, 3] carry minted Zenodo DOIs; the remaining Tier-1 and Tier-2 companions cited above are pre-DOI.

Result. The page-long $\mathcal{L}_{\mathrm{SM}}$ is not assumed. In machine-checked Lean (zero `sorry`), we prove a constructive sector-by-sector bridge: gauge kinetics from the abelian octonion kinetic density split by the force-sector map; fermion minimal coupling from the triality-summed source slot; Higgs kinetic and potential from the octonion-scalar norm evaluated on the dynamic vacuum expectation value; Yukawa weights from the TUFT/Hopf mass chart at the chosen now slice. The headline theorem packages all five sector equalities simultaneously. The topological suppression in the electroweak mass chart factorises as $\kappa_6(\xi, \Phi, t) = \eta_{\mathrm{local}}(\xi)\,\gamma\,C_2(\xi, \Phi, t)$: matter fraction $\eta_{\mathrm{local}} = \eta_{\mathrm{paper}}\,\Omega_k(\xi)$ at lock-in normalization, overlap $\gamma = 2/5$, and second-order lapse concentration C_2 from δ-corrected Rindler detuning at the readout point (proved $C_2 = 56/45$ at ξ_{lock} with $\Phi = t = 0$). Because the effective vev and κ_6 are explicit functions of the selected now slice—via inner–outer Casimir balance, the curvature primitive, and HQVM lapse data—the entire Lagrangian at any epoch is determined geometrically once that slice is chosen. There are no remaining free parameters in the conventional sense: no fitted Yukawa matrix, no tuned C_2, and no simultaneous PDG mass solve. The combinatorial lattice fixes $\alpha = 3/5$, $\gamma = 2/5$, $\alpha_{\mathrm{GUT}} = 1/42$; the only operational choice is a consistent “now” slice (ξ, Φ, t) on the temperature ladder. PDG centrals and CODATA couplings are comparison layers only. Reader-friendly Lean anchors are in Appendix A.

Patch-ontology reader contract. Per the patch-theory reader contract below, the discrete patch layer is the ontology. Smooth-manifold textbook notation for $\mathcal{L}_{\mathrm{SM}}$ is a calculation-approximation rewrite for cross-citation only—not a foundational target, not a mesh-refinement programme, and not required for validity of the discrete bridge. At the load-bearing steps (shell index m, resonance ladder, κ_6 factorisation, and the rationals above) a literal continuum rewrite would erase the discrete index doing the predictive work.

Status. What is proved: the five sector equalities, the total-action split, and Yukawa linearity over flavours, all under the dynamic inner–outer Casimir scale. The theory now carries zero free parameters beyond the discrete lattice data; the only operational choice is the identification of a consistent “now” slice on the temperature ladder. What is named explicitly and *not* claimed: anomaly cancellation, measure choice, and the full non-abelian holonomy upgrade (scaffolded upstream but not closed here).

Patch QFT and discrete numerics (reader contract). HQIV (Horizon-Quantized Informational Vacuum) is formulated as a *patch theory*: quantum fields, actions, and physical readouts are attached to discrete patches—shell indices $m \in \mathbb{N}$, finite spacetime index types (e.g. $\mathrm{Fin}\,4$), and horizon-limited charts—rather than to a hypothetical fundamental smooth spacetime obtained as a mesh-refinement or ultraviolet limit. Accordingly, when this text refers to “QFT,” it means *patch QFT*: patching rules, finite measurement ledgers, and variational identities on the discrete spine; it is not the claim that the theory is *defined* by recovering ordinary continuum quantum field theory through such a limit. The numbers that admit the sharpest statements—mass and coupling

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ladders, curvature ratios, shell thermodynamics, and rational witnesses in `HQIV_LEAN`—are objects of *discrete mathematics* on $\mathbb{N}$ (combinatorial growth, cumulative simplex ledgers) together with the ordered field $\mathbb{R}$ as used in the proof assistant for analysis, bounds, and phenomenological display. Smooth-manifold or continuum field notation, when it appears, is a *translation layer* for comparison with standard physics literature; it does not supersede the discrete definitions.

A continuum is not required, and in places would destroy these results. The discrete patch layer is *not* a numerical approximation to a more fundamental continuum theory awaiting future construction; it is the ontology. Where the literature treats "the continuum limit" as the goal, HQIV treats it as a *calculation approximation* for comparison with standard formulas. The continuum form is an optional convenience for comparison; the proved discrete statement is what carries the physics. In several load-bearing places—the curvature imprint $\delta_E(m)$ at shell m, the harmonic ladder masses $m_\tau \to m_\mu \to m_e$, the lock-in vev $v = \sqrt{\eta_{\text{paper}}\,\Omega_k(m_{\text{lockin}}; m_{\text{lockin}})}$, the proton anchor at $m = \text{referenceM} = 4$, the baryon-asymmetry $\eta \propto 6^7\sqrt{3}/(m+1)$, and the rational coefficients $\alpha = 3/5$, $\gamma = 2/5$, $\alpha_{\text{GUT}} = 1/42$—taking a literal $m \to \infty$ or sub-Planck continuum limit would *erase* the discrete index that is doing the predictive work. Continuum forms of these objects are therefore not preferred refinements; they are degenerate, information-losing readouts of the same patch data.

An exception: $\mathfrak{so}(8)$ as a de-facto continuum algebra from a discrete construction. Principled exception to the discrete-only stance: the closed Lie algebra $\mathfrak{so}(8) \supseteq G_2 \cup \{\Delta\}$ generated from the Fano octonion left-multiplication table and the phase-lift generator $\Delta \in (\mathrm{e}_1, \mathrm{e}_7)$. Although $\mathfrak{so}(8)$ is a continuous (28-dimensional real) Lie algebra, it is built here entirely from *discrete* ingredients (the seven imaginary octonion units, a finite Fano table, and one $\pi/2$ rotation), and its closure to 28 dimensions is a finite linear-algebra certificate (see the `Hqiv.SO8Closure` / `Hqiv.SO8ClosureInterface` stack). The Lie-group structure is therefore a *de-facto continuum algebra obtained from a discrete construction*: continuous in parameter space, but with no continuum spacetime, no continuum measure, and no UV limit attached. When this manuscript or its companions write $\exp(t\,\Delta) \in \mathrm{SO}(8)$ or treat $\mathfrak{so}(8)$ rotations as smooth, those are statements internal to a finite-dimensional Lie group whose generators are certified by discrete data; they are *not* a continuum-spacetime commitment and do not require a continuum-QFT scaffolding.

1. Scope and claim discipline

1.1. **No free parameters; only now-slice selection.** With the inner–outer Casimir dynamics on the same Hopf-shell carrier and the curvature primitive along the physical-temperature ladder in place, the construction contains no free parameters in the conventional sense. The combinatorial data of the lattice fix the dimensionless rationals $\alpha = 3/5$, $\gamma = 2/5$ and $\alpha_{\text{GUT}} = 1/42$ once and for all. The only operational choice in any given solve or phenomenological export is the selection of a consistent "now" slice on the temperature ladder. Once that slice is identified, the dynamic scale determines the effective vacuum expectation value at that epoch, the resonance ladder supplies the Yukawa weights, and the sector projections of the discrete octonion action yield the full Lagrangian. Different choices of now slice generate different effective spectra and couplings; the geometry supplies the explicit temperature evolution that relates them. The proton lock-in at $\text{referenceM} = 4$ remains a natural calibration point for the hadronic sector, but it is no longer required for the leptonic or gauge sectors.

1.2. **Mass-scale closure: local matter fraction and lapse concentration.** The TUFT/Hopf bridge of [8] supplies the electroweak mass chart $\Lambda_{\text{Hopf}}(\xi) = \sqrt{2\pi}\,v(\xi)\,\kappa_6(\xi, \Phi, t)$. The former fitted slot $C_2 \approx 1.135$ is replaced in Lean by a derived readout (zero `sorry` in `Hqiv/Physics/HopfShellBeltramiMassBridge.lean`):

$$\kappa_6(\xi, \Phi, t) = \eta_{\text{local}}(\xi)\,\gamma_{\text{HQIV}}\,C_2(\xi, \Phi, t), \qquad \eta_{\text{local}}(\xi) = \eta_{\text{paper}}\,\frac{\Omega_k(\xi)}{\Omega_k(\xi_{\text{lock}})}. \tag{1}$$

At integer shells m with horizon referenceM, the local matter fraction agrees with the baryogenesis witness $\eta_{\mathrm{local}}(\xi) = \eta(m; \mathrm{referenceM})$ (local η on shells). The overlap $\gamma_{\mathrm{HQIV}} = 2/5$ is the proved horizon-split coefficient (horizon-split γ identity). The second-order factor is *lapse concentration* at the readout point:

$$C_2(\xi, \Phi, t) = \frac{1+\gamma}{2} \, \frac{\texttt{rindlerDenWithDelta}\big(\lambda \,\mathrm{obs}(\xi, \Phi, t),\, \mathrm{referenceM}\big)}{\texttt{rindlerDenShared}(\mathrm{referenceM})}, \tag{2}$$

with $\lambda = c_{\mathrm{rindler}} = \gamma/2$ and $\mathrm{obs}(\xi, \Phi, t) = \Theta_{\mathrm{local}}(\xi)\,(1+\gamma) + (N-1)$ for $N = \texttt{HQVM\_lapse}(\Phi, \varphi(\xi), t)$. This packages the same δ-corrected detuning observable used in [5] and the global-detuning module: matter curvature is corrected *locally* by $\Omega_k(\xi)$; observer concentration enters through C_2, not through a hidden Yukawa fit.

At lock-in ($\xi_{\mathrm{lock}} = 5$, $\Phi = t = 0$, referenceM $= 4$) Lean proves $C_2 = 56/45$ (lock-in lapse concentration). The pre-closure regression constant $1.135136\ldots$ is retained only as `tuftHopfKappa6SecondOrderCorrectionLegacy` for chart comparison. The physical κ_6 used by neutrino and lepton readouts is `tuftHopfKappa6AtXi` $\xi\,\Phi\,t$; at lock-in it is `tuftHopfKappa6` (κ_6 at horizon).

For the SM Lagrangian assembly, the lock-in vev (lock-in vev) remains the weak-sector scalar anchor $\sqrt{\eta_{\mathrm{paper}}}$ when $\Omega_k = 1$ at lock-in, while charged-lepton masses at epoch ξ follow the dynamic vev/TUFT chart of [8]. The five sector equalities of Theorem 6.3 are unchanged; what changes is that the mass *normalization* feeding $\mathcal{L}_Y$ is no longer a separate fit layer once (ξ, Φ, t) is fixed.

1.3. **Predecessor: discrete O–Maxwell action.** [5] packages the discrete octonion-indexed gauge potential ${A^a}_\mu$ (channels $a \in \mathrm{Fin}\,8$, spacetime indices $\mu, \nu \in \mathrm{Fin}\,4$), field strength ${F^a}_{\mu\nu}(A) = {A^a}_\nu - {A^a}_\mu$ (the field-strength constructor), the kinetic density $L_O^{\mathrm{kin}}(A)$ (the octonion kinetic density), the source interaction $L_O^{\mathrm{src}}(J_{\mathrm{src}}, A)$ (the octonion source coupling), and the ϕ-coupling slot $L_O^{\phi}(A, \phi_{\mathrm{val}})$ (the scalar ϕ-coupling slot) into the single Lagrangian density (the general O-Maxwell Lagrangian density)

$$\mathcal{L}_O[J_{\mathrm{src}}](A, \phi_{\mathrm{val}}) = L_O^{\mathrm{kin}}(A) + 4\pi\, L_O^{\mathrm{src}}(J_{\mathrm{src}}, A) + L_O^{\phi}(A, \phi_{\mathrm{val}}).$$

The gravitational constraint scalar $S_{\mathrm{HQVM\,grav}}$ (the HQVM gravitational constraint) vanishes iff the HQVM Friedmann equation holds (the Friedmann equivalence theorem). The total cell (the HQIV total action) is $\mathcal{L}_O + S_{\mathrm{HQVM\,grav}}$. [1] adds the charged-lepton resonance ladder (the two resonance steps), the τ-Planck anchor (the tau Planck mass), and the generation-level mass functional (the geometric SM mass functional) that ties an SM flavour label to a Planck-unit rest mass with no PDG input. [6] and the SM embedding module fix the $\mathrm{SU}(2)_L$ and hypercharge generators inside the certified $G_2 \cup \{\Delta\} \Rightarrow \mathfrak{so}(8)$ closure, exhaust the standard branching rules on 8_s, and certify the three triality irreps that we read as generation labels.

1.4. **What this paper adds.** The new Lean module is the Standard-Model Lagrangian bridge (zero `sorry`, builds clean under `lake build`); its principal contributions are:

- **Sector splitting of the abelian octonion kinetic** (Sect. 2): the three-sector kinetic split writes $L_O^{\mathrm{kin}} = L_{O,\mathrm{EM}}^{\mathrm{kin}} + L_{O,\mathrm{Weak}}^{\mathrm{kin}} + L_{O,\mathrm{Strong}}^{\mathrm{kin}}$ under the explicit force-sector assignment (the force-sector map) from the Forces module; sector aggregates project to the textbook B, W, G kinetic blocks; the SM gauge total equals L_O^{kin} exactly (the gauge–kinetic equality).
- **Triality-summed fermion coupling** (Sect. 3): per-generation fermion currents on the 8_s carrier are summed by two applications of the proved J-linearity (the source-coupling additivity lemma); the three-generation equality packages the result.
- **Higgs as the octonion scalar** (Sect. 4): the Higgs kinetic block is the per-direction sum of the octonion scalar norm; the symmetry-breaking potential is the existing Higgs potential with $\lambda \mapsto \lambda_{\mathrm{eff}}$ (the effective quartic) and $v \mapsto$ `lockinVev` (the lock-in vev), expanded to the textbook quartic form (the expanded Higgs potential identity).

- **Mass-scale closure** (Sect. 1.2): $\kappa_6(\xi, \Phi, t) = \eta_{\text{local}}(\xi)\,\gamma\,C_2(\xi, \Phi, t)$ with local η, proved lapse concentration at lock-in, and κ_6 factorisation witness.
- **Yukawa from the TUFT mass chart** (Sect. 5): $y_f := \sqrt{2}\,m_f(\xi)/v(\xi)$ with m_f from the Hopf/Beltrami scalar at the now slice; the identity $y_f v/\sqrt{2} = m_f$ is proved (the Yukawa–mass identity).
- **Assembled SM Lagrangian** (Sect. 6): the SM Lagrangian record and its from-HQIV constructor package the five sector pieces; the headline theorem (the SM-from-HQIV discrete-action theorem) certifies the five sector equalities simultaneously; the total-action equality (the total-action split theorem) exhibits the SM gauge + fermion + ϕ-coupling + Friedmann split of the HQIV total cell when $J_{\text{src}} = \sum_{\text{gen}} J(\text{gen})$.
- **Parameter witness** (Sect. 7): the parameter witness certifies $\alpha = 3/5$, $\gamma = 2/5$, $\alpha_{\text{GUT}} = 1/42$; the κ_6 factorisation replaces every former fitted second-order slot—not a simultaneous PDG fit set.

1.5. **What is and isn't proved.** This module is a *structural bridge*, not a re-derivation of textbook field equations:

- **Proved (Lean).** The five sector equalities of the SM-from-HQIV discrete-action theorem; the total-action split of the total-action split theorem; the parameter witness; linearity of the Yukawa sum (the Yukawa sum identity).
- **Black-box ingredients.** The HQIV kinetic, source, ϕ-coupling, and gravitational pieces, together with the Higgs potential, lock-in vev, geometric mass functional, effective quartic, and force-sector map, are imported from upstream modules; all `sm_*_from_HQIV` theorems are tautological equalities. Any smooth-manifold translation is a comparison layer only. Load-bearing predictions of [5] live on the discrete spine.
- **Symbolic kinetic terms.** Fermion and Higgs kinetic shadows remain on $\Phi : \text{Fin}\,8 \to \mathbb{R}$; no BRST quantisation here.
- **Non-abelian holonomy.** Per-sector kinetic uses abelian $F^a{}_{\mu\nu} = A^a{}_\nu - A^a{}_\mu$; the weak commutator scaffold is upstream and not closed in this module.
- **Single witness, no PDG fit.** Yukawa weights come from the resonance ladder and τ-Planck anchor on the discrete spine; the Higgs vev is the lock-in v above. Under the default proton lock-in witness of [1, 3], PDG masses are comparison targets, not simultaneous solve anchors.

2. Sector projection of the gauge kinetic block

The HQIV abelian octonion kinetic density (the octonion kinetic density of [5]),

$$L_O^{\text{kin}}(A) = -\tfrac{1}{4} \sum_{a \in \text{Fin}\,8} \sum_{\mu,\nu \in \text{Fin}\,4} \tfrac{(F^a{}_{\mu\nu}(A))^2}{2},$$

splits into three sector pieces. The force-sector assignment is the O-component map forceSector(a) with forceSector$(0) = \text{EM}$, forceSector$(a) = \text{Weak}$ for $a \in \{1, 2, 3\}$, and forceSector$(a) = \text{Strong}$ for $a \in \{4, 5, 6, 7\}$ (the force-sector map), fixed in the Forces module; component 0 in the quaternionic subalgebra reduces to classic Maxwell (the zeroth-component is EM lemma), the three remaining quaternionic components carry the weak-like sector, and the four octonion-extension components carry the colour-like sector.

Definition 2.1 (Sector projections of L_O^{kin})**.** *For $s \in \{EM, Weak, Strong\}$ and a potential A,*

$$L_{O,s}^{\text{kin}}(A) := -\tfrac{1}{4} \sum_{a \in \text{Fin}\,8} \chi_s(a) \sum_{\mu,\nu \in \text{Fin}\,4} \tfrac{(F^a{}_{\mu\nu}(A))^2}{2},$$

where $\chi_s(a) = 1$ *if* $\mathrm{forceSector}(a) = s$ *and zero otherwise (the sector-projected kinetic density). Aggregating:*

$$\mathcal{L}_B(A) := L^{\mathrm{kin}}_{O,\mathrm{EM}}(A), \qquad \mathcal{L}_W(A) := L^{\mathrm{kin}}_{O,\mathrm{Weak}}(A), \qquad \mathcal{L}_G(A) := L^{\mathrm{kin}}_{O,\mathrm{Strong}}(A)$$

(the hypercharge kinetic, weak kinetic, and strong kinetic aggregates).

Theorem 2.2 (Three-sector splitting, Lean). *For every* A,

$$L^{\mathrm{kin}}_O(A) = L^{\mathrm{kin}}_{O,\mathrm{EM}}(A) + L^{\mathrm{kin}}_{O,\mathrm{Weak}}(A) + L^{\mathrm{kin}}_{O,\mathrm{Strong}}(A).$$

This is the three-sector kinetic split.

Sketch (Lean tactic translation). For each octonion channel $a \in \mathrm{Fin}\,8$, exactly one of the three indicator functions $\chi_{\mathrm{EM}}(a), \chi_{\mathrm{Weak}}(a), \chi_{\mathrm{Strong}}(a)$ is one and the other two vanish (case split on $\mathrm{forceSector}(a)$). Summing the channel-wise identity over a and applying $-\frac{1}{4}\cdot$ yields the claim. Lean discharges the finite case split with `cases` + `simp` and the global sum with `Finset.sum_add_distrib`. □

Corollary 2.3 (SM gauge kinetic equals HQIV octonion kinetic). $\mathcal{L}_B(A)+\mathcal{L}_W(A)+\mathcal{L}_G(A) = L^{\mathrm{kin}}_O(A)$ *(the gauge–kinetic equality).*

Reading. The textbook SM gauge kinetic block $\mathcal{L}_{\mathrm{gauge}} = -\frac{1}{4}(B_{\mu\nu}B^{\mu\nu} + W^I_{\mu\nu}W^{I\mu\nu} + G^a_{\mu\nu}G^{a\mu\nu})$ is the sector-projected aggregate of the single HQIV object L^{kin}_O when the force-sector map is the identification fixed by the algebra (colour-preferred axis and hypercharge block of the SM embedding module). No new coupling, no new field strength: the gauge-kinetic page of $\mathcal{L}_{\mathrm{SM}}$ is one discrete object with a sector partition.

3. Fermion sector: minimal coupling and three generations

The textbook block $i\bar{\psi}\gamma^\mu D_\mu\psi$ summed over one chiral SM generation is supported on the 8-dimensional left-handed 8_s carrier of the SM embedding module, plus 8 right-handed components in 8_c (see the branching rule on 8_s documented there). At the abelian level needed for the present bridge, the Dirac bilinear is encoded as a current $J : \mathrm{Fin}\,8 \to \mathrm{Fin}\,4 \to \mathbb{R}$ on the 8s carrier, and the minimal coupling reads off the proved $J \cdot A$ slot L^{src}_O.

Definition 3.1 (Per-generation minimal coupling). *For a triality generation label* $\mathrm{gen} \in$ *`So8RepIndex` and a generation current* J_{gen},

$$L^{\mathrm{SM}}_{\mathrm{fermion}}(\mathrm{gen}, J_{\mathrm{gen}}, A) := L^{\mathrm{src}}_O(J_{\mathrm{gen}}, A)$$

(the per-generation minimal coupling). The three SM generations correspond to the three Spin(8) irreps $8_v, 8_{s^+}, 8_{s^-}$ *counted by the three-generations-from-triality theorem* (card *`So8RepIndex`* $= 3$*).*

Theorem 3.2 (Triality-summed fermion coupling, Lean). *For any triality-indexed family of currents* J : *`So8RepIndex`* $\to (\mathrm{Fin}\,8 \to \mathrm{Fin}\,4 \to \mathbb{R})$,

$$\sum_{\mathrm{gen}\in\{8_v,8_{s^+},8_{s^-}\}} L^{\mathrm{SM}}_{\mathrm{fermion}}(\mathrm{gen}, J(\mathrm{gen}), A) = L^{\mathrm{src}}_O\left(\sum_{\mathrm{gen}} J(\mathrm{gen}),\ \mathrm{A}\right).$$

This is the three-generation equality; its proof is two applications of the source-coupling additivity lemma (Lemma 3.1 of [5]).

Remark 3.3 (8_v, 8_{s^+}, 8_{s^-} vs. generation labelling). *The SM–GR unification module assigns generation* $1 \to 8_v$*, generation* $2 \to 8_{s^+}$*, generation* $3 \to 8_{s^-}$ *via the generation index map; this is a representation-counting fact, not a mass theorem. The mass hierarchy itself is read off the charged-lepton resonance module via the resonance-product factor (Section 5 below).*

Reading. The textbook fermion sector $\mathcal{L}_{\text{fermion}} = \sum_{\text{gen}} i\bar{\psi}_{\text{gen}}\gamma^\mu D_\mu \psi_{\text{gen}}$ is, at the bookkeeping level relevant to the discrete action, the abelian source coupling for the sum of the three triality-indexed currents on the 8_s carrier. The covariant derivative D_μ encodes the same gauge connection A that supplies the sector kinetic block of Section 2; no additional gauge structure is needed.

4. Higgs sector: octonion scalar, vev, and quartic

The HQIV scalar lives on the same $\mathrm{Fin}\,8 \to \mathbb{R}$ carrier as the gauge potential (the weak/Higgs scaffold); the Higgs kinetic shadow of the textbook $(D_\mu\Phi)^\dagger(D^\mu\Phi)$ is read off the octonion scalar norm, summed over spacetime directions; the symmetry-breaking potential is the Higgs potential at the calibrated lock-in vev.

Definition 4.1 (Higgs sector projection). *For a per-direction scalar slot $\Phi_{\text{dir}} : \mathrm{Fin}\,4 \to (\mathrm{Fin}\,8 \to \mathbb{R})$,*

$$\mathcal{L}^{\text{kin}}_{\text{Higgs}}(\Phi_{\text{dir}}) := \sum_{\mu\in\mathrm{Fin}\,4} \mathrm{scalarNormSq}(\Phi_{\text{dir}}(\mu))$$

(the SM Higgs kinetic shadow). For a scalar configuration Φ,

$$\mathcal{L}^{\text{pot}}_{\text{Higgs}}(\Phi) := \texttt{higgsPotential}(\lambda_{\text{eff}},\ \texttt{lockinVev},\ \Phi) = \lambda_{\text{eff}}\left(|\Phi|^2 - \texttt{lockinVev}^2\right)^2$$

(the SM Higgs potential shadow), with λ_{eff} from the SM–GR unification module and the lock-in vev from the weak/Higgs scaffold,

$$\texttt{lockinVev} := \sqrt{\eta_{\text{paper}}\,\Omega_k(m_{\text{lockin}}; m_{\text{lockin}})}.$$

Theorem 4.2 (Textbook expanded form, Lean).

$$\mathcal{L}^{\text{pot}}_{\text{Higgs}}(\Phi) = \lambda_{\text{eff}}\,|\Phi|^4 - 2\lambda_{\text{eff}}\ \texttt{lockinVev}^2\,|\Phi|^2 + \lambda_{\text{eff}}\ \texttt{lockinVev}^4.$$

This is the expanded Higgs potential identity (proved by `unfold` *plus* `ring`*).*

Remark 4.3 (Lock-in calibration of the vev). *The calibration $\texttt{lockinVev}^2 = \eta_{\text{paper}}$ when $\Omega_k(m_{\text{lockin}}; m_{\text{lockin}}) = 1$ is the lock-in vev squared identity; the curvature ratio at lock-in is the lock-in Ω_k calibration. The lock-in vev is therefore* not *a fitted constant: it is the square root of the proved baryogenesis η-anchor, packaged once and reused unchanged in the Higgs sector.*

Reading. The textbook Higgs page becomes, in HQIV notation, the diagonal scalar-norm sum (kinetic) plus the expanded Higgs potential with $(\lambda, v^2) = (\lambda_{\text{eff}}, \eta_{\text{paper}})$. The constant $\lambda_{\text{eff}}\,v^4$ is a global vacuum offset that does not enter equations of motion.

5. Yukawa sector from the resonance ladder

The Yukawa block of $\mathcal{L}_{\text{SM}}$ is $\mathcal{L}_Y = -\sum_f y_f\,\bar{f}\Phi f + \text{h.c.}$. In HQIV the Yukawa coupling for each SM flavour is the textbook combination $y_f = \sqrt{2}\,m_f(\xi)/v(\xi)$ at the selected now slice, with $v(\xi) = \texttt{tuftVevAtXi}(\xi)$ from the dynamic Casimir chart and $m_f(\xi)$ from the TUFT Hopf/Beltrami scalar ([8], TUFT lepton mass at ξ). At lock-in the legacy geometric ladder of [1] remains available as a diagnostic, but the physical chart uses $\Lambda_{\text{Hopf}} = \sqrt{2\pi}\,v\,\kappa_6(\xi, \Phi, t)$ with κ_6 from (1)—not a separate Yukawa fit.

Definition 5.1 (SM Yukawa coupling). *For an SM flavour label $\ell \in$* `SMMassLabel` *(the SM mass-label type),*

$$y_{\text{SM}}(\ell) := \frac{\sqrt{2}\,m^{\text{geom}}_\ell}{\texttt{lockinVev}}$$

(the SM Yukawa coupling). For a symbolic Dirac bilinear $B :$ `SMMassLabel` *$\to \mathbb{R}$,*

$$\mathcal{L}^{\text{SM}}_{\text{Yukawa}}(\ell, B) := -\,y_{\text{SM}}(\ell)\ B(\ell)$$

(the per-flavour Yukawa term); the all-flavour aggregate is the list sum over twelve SM flavours (the all-flavours list, length certified by the all-flavours length lemma).

Theorem 5.2 (Yukawa identity, Lean)**.** *For every SM flavour ℓ and every $v = \texttt{lockinVev} \neq 0$,*

$$\frac{y_{\rm SM}(\ell)\, v}{\sqrt{2}} = m_\ell^{\rm geom} = m_\tau^{\rm Pl} \;/\; \mathrm{resonanceProduct}(\mathrm{genIndex}(\ell)).$$

This is the pair comprising the Yukawa–mass identity and the Yukawa resonance witness.

Theorem 5.3 (Yukawa linearity over flavours, Lean)**.** *For any flavour list L and bilinear assignment B,*

$$\sum_{\ell \in L} \mathcal{L}^{\rm SM}_{\rm Yukawa}(\ell, B(\ell)) = -\sum_{\ell \in L} y_{\rm SM}(\ell)\, B(\ell).$$

This is the Yukawa sum identity.

Remark 5.4 (No fitted Yukawa matrix)**.** *The textbook SM stores $\mathcal{O}(20)$ real Yukawa and mixing parameters fitted from experiment. In HQIV these collapse to: three lattice rationals, one now slice (ξ, Φ, t), and the proved factorisation (1)–(2). Laboratory masses enter only as comparison targets. The baryogenesis anchor $\eta_{\rm paper}$ is the same surplus wired into $\eta_{\rm local}(\xi)$ and into* `lockinVev`*; the dynamic bulk integrator (*`scripts/hqiv_dynamic_bulk_bbn.py`*) propagates $\eta_{\rm local}$ shell-by-shell and reports emergent lock-in without re-fitting η at each step.*

Reading. The textbook Yukawa "page" $-y_e \bar{L}\Phi e_R - y_u \bar{Q}\tilde{\Phi} u_R - y_d \bar{Q}\Phi d_R + \text{h.c.}$ (three generations) becomes, in HQIV notation, a list-sum over twelve flavours each weighted by $\sqrt{2}\, m_\tau^{\rm Pl}/(\texttt{lockinVev} \cdot \mathrm{resonanceProduct})$. *Three* numbers $(m_\tau^{\rm Pl},\, k_{\tau\mu},\, k_{\mu e})$ fix the entire Yukawa sector.

6. Assembled SM Lagrangian and headline theorems

Definition 6.1 (SM Lagrangian record (Lean))**.** *The SM Lagrangian record carries five real-valued fields $(\mathcal{L}_{\rm gauge}, \mathcal{L}_{\rm fermion}, \mathcal{L}^{\rm kin}_{\rm Higgs}, \mathcal{L}^{\rm pot}_{\rm Higgs}, \mathcal{L}_Y)$. The total is*

$$\mathcal{L}^{\rm total}_{\rm SM} := \mathcal{L}_{\rm gauge} + \mathcal{L}_{\rm fermion} + \mathcal{L}^{\rm kin}_{\rm Higgs} - \mathcal{L}^{\rm pot}_{\rm Higgs} + \mathcal{L}_Y$$

(the SM Lagrangian total).

Definition 6.2 (SM Lagrangian from HQIV (Lean))**.** *Given an HQIV gauge potential A, a triality-indexed current family J, an octonion scalar Φ, a per-direction scalar slot $\Phi_{\rm dir}$, and a flavour bilinear assignment B, the from-HQIV constructor sets*

$$\mathcal{L}_{\rm gauge} = L_O^{\rm kin}(A), \quad \mathcal{L}_{\rm fermion} = \sum_{\rm gen} L_O^{\rm src}(J(\mathrm{gen}), \mathrm{A}),$$

$$\mathcal{L}^{\rm kin}_{\rm Higgs} = \sum_\mu \mathrm{scalarNormSq}(\Phi_{\rm dir}(\mu)), \quad \mathcal{L}^{\rm pot}_{\rm Higgs} = \texttt{higgsPotential}(\lambda_{\rm eff}, \texttt{lockinVev}, \Phi),$$

$$\mathcal{L}_Y = \sum_{\ell \in \texttt{all\_SM\_flavours}} \mathcal{L}^{\rm SM}_{\rm Yukawa}(\ell, B(\ell)).$$

Theorem 6.3 (Headline: SM Lagrangian = HQIV discrete action)**.** *For every $(A, J, \Phi, \Phi_{\rm dir}, B)$, writing L for the from-HQIV record at those arguments,*

(1) $\mathcal{L}_{\rm gauge}(L) = L_O^{\rm kin}(A)$;
(2) $\mathcal{L}_{\rm fermion}(L) = L_O^{\rm src}\big(J^{\rm tot}_{\rm SM}(J), A\big)$, where $J^{\rm tot}_{\rm SM}$ is the triality-summed SM current;
(3) $\mathcal{L}^{\rm kin}_{\rm Higgs}(L) = \sum_\mu \mathrm{scalarNormSq}(\Phi_{\rm dir}(\mu))$;
(4) $\mathcal{L}^{\rm pot}_{\rm Higgs}(L) = \texttt{higgsPotential}(\lambda_{\rm eff}, \texttt{lockinVev}, \Phi)$;
(5) $\mathcal{L}_Y(L)$ is the twelve-flavour Yukawa sum with weights $\sqrt{2}\, m_\ell^{\rm geom} / \texttt{lockinVev}$.

This is the SM-from-HQIV discrete-action theorem.

Theorem 6.4 (Total HQIV action split, Lean)**.** *With* $J^{\mathrm{tot}}_{\mathrm{SM}}(J)$ *as in Theorem 6.3,*

$$\mathcal{S}^{\mathrm{total}}_{\mathrm{HQIV}}\big(J^{\mathrm{tot}}_{\mathrm{SM}}(J), A, \phi_{\mathrm{val}}, \rho_m, \rho_r\big) = \Big(L^{\mathrm{kin}}_O(A) + 4\pi \sum_{\mathrm{gen}} L^{\mathrm{src}}_O(J(\mathrm{gen}), A) + L^{\phi}_O(A, \phi_{\mathrm{val}})\Big) + S_{\mathrm{HQVM\,grav}}(\phi_{\mathrm{val}}, \rho_m, \rho_r).$$

This is the total-action split theorem; its proof uses two invocations of the source-coupling additivity lemma, then **`ring`** *on the residue.*

Reading. Theorem 6.4 exposes the dictionary

$$\boxed{\mathcal{S}_{\mathrm{HQIV}} = \mathcal{L}^{\mathrm{SM}}_{\mathrm{gauge}} + \frac{1}{4\pi}\mathcal{L}^{\mathrm{SM}}_{\mathrm{fermion}} + \mathcal{L}^{\mathrm{SM}}_{\mathrm{Higgs\ cross}} + S_{\mathrm{HQVM\,grav}}}$$

between the page-long SM Lagrangian and the HQIV total cell. The $1/(4\pi)$ factor is the discrete normalization from the Maxwell-source convention of the general O-Maxwell Lagrangian; the Higgs-cross label captures the ϕ-coupling remnant once Φ is collapsed to the lattice scalar at the electroweak shell (shell index 5 in the SM–GR module).

7. Parameter accounting and witnesses

The full from-HQIV construction consumes the following *derived* discrete inputs. None is a simultaneous PDG fit; laboratory comparison uses the proton lock-in witness discipline of [1, 3].

Table 1. Inputs to the HQIV-derived SM Lagrangian after the κ_6 closure. None is a simultaneous PDG fit; the only operational choice is the now slice (ξ, Φ, t).

Slot	HQIV source	Appendix link
$\alpha = 3/5$	Lattice stars-and-bars	lattice α identity
$\gamma = 2/5$	Horizon split $1-\alpha$	horizon-split γ identity
$\alpha_{\mathrm{GUT}} = 1/42$	Cube $\times$ octonion channels	GUT coupling identity
$\eta_{\mathrm{local}}(\xi)$	$\eta_{\mathrm{paper}}\,\Omega_k(\xi)$	local matter fraction
$C_2(\xi, \Phi, t)$	Lapse concentration (2)	lock-in $C_2 = 56/45$
κ_6	$\eta_{\mathrm{local}}\,\gamma\,C_2$	factorisation
$v(\xi)$, $m_f(\xi)$	Dynamic Casimir + TUFT scalar	TUFT masses at ξ
λ_{eff}	EW-shell quartic	effective quartic
`lockinVev`	$\sqrt{\eta_{\mathrm{paper}}}$ at $\Omega_k = 1$	lock-in vev

Theorem 7.1 (Parameter witness, Lean)**.** $\alpha = 3/5$, $\gamma_{\mathrm{HQIV}} = 2/5$, $\alpha_{\mathrm{GUT}} = 1/42$ *simultaneously. This is the parameter witness, discharged by the three rational identities listed in the table.*

Theorem 7.2 (Yukawa resonance witness, Lean)**.** *For every SM flavour ℓ, when* `lockinVev` $\neq 0$,

$$\frac{y_{\mathrm{SM}}(\ell)\ \mathtt{lockinVev}}{\sqrt{2}} = m^{\mathrm{Pl}}_{\tau} \,/\, \mathrm{resonanceProduct}(\mathrm{genIndex}(\ell)).$$

This is the Yukawa resonance witness.

Remark 7.3 (Comparison with textbook parameter count)**.** *The textbook SM has $\mathcal{O}(20)$ fitted real parameters. The HQIV bridge compresses them to: three proved lattice rationals, one now slice (ξ, Φ, t), and the derived κ_6 factorisation. Gauge couplings trace to α_{GUT} and the ϕ-correction of* [5]*; charged-fermion masses trace to the TUFT chart; (μ^2, λ) trace to $(\eta_{\mathrm{paper}}, \lambda_{\mathrm{eff}})$ at lock-in. The dynamic baryogenesis integrator recovers η_{paper} at lock-in to $\sim 1\%$ on the Ω_b path without injecting a separate η knob per shell.*

8. Outlook: what comes next

The structural bridge of Theorem 6.3 is exact at the bookkeeping level. To translate it to textbook smooth-manifold notation (a calculation-approximation layer for comparison with [10, 11], *not* a foundational completion of HQIV), three Tier-III tasks remain (ordering consistent with [5]):

(1) **Variational fermion sector.** Promote the symbolic current to a typed Dirac bilinear on the $\mathrm{Cl}(0,6)$ spinor carrier already present in the library; the abelian source coupling of Section 3 becomes the textbook minimal coupling.
(2) **Non-abelian gauge kinetic.** Replace abelian ${F^a}_{\mu\nu}$ in the weak and strong sectors by the non-abelian weak field-strength scaffold and the $\mathrm{SU}(3)_c$ chart Lie law; the reduction theorem upstream guarantees collapse to the abelian core when the commutator vanishes.
(3) **Higgs covariant derivative.** Wire full gauge connection data into the fermion and Higgs slots; verify the kinetic shadow reduces to the scalar-norm sum in the abelian limit.

None of these tasks requires new physics input. The discrete constants, κ_6 factorisation, dynamic vev, and mass chart are already proved; smooth-manifold form is a translation layer only—not a continuum-completion programme.

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Appendix A. Lean module index

The Lean development lives at `https://github.com/HQIV/hqiv-lean` [9]. Reader-friendly anchors in the body link here.

Standard-Model Lagrangian bridge.:
Headline module: five sector equalities and parameter witness. `Hqiv/Physics/StandardModelLagrangianFromDiscreteAction.lean`.
O-Maxwell action module.:
Discrete gauge action upstream. `Hqiv/Physics/Action.lean`.
Field-strength constructor.:
${F^a}_{\mu\nu}(A) = {A^a}_\nu - {A^a}_\mu$. `F_from_A`.
Octonion kinetic density.:
L_O^{kin}. `L_O_kinetic`.
Octonion source coupling.:
L_O^{src}. `L_O_source_general`.
Scalar ϕ-coupling slot.:
`L_O_phi_coupling`.
General O-Maxwell Lagrangian.:
`L_O_Maxwell_general`.
HQVM gravitational constraint.:
`S_HQVM_grav`.
Friedmann equivalence.:
`S_HQVM_grav_zero_iff_Friedmann`.
HQIV total action.:
`action_total_general`.
Forces module.:
`Hqiv/Physics/Forces.lean`.

Force-sector map.:
`O_component_to_sector`.
Zeroth component is EM.:
`O_component_zero_is_EM`.
Sector-projected kinetic.:
`L_O_kinetic_sector`.
Three-sector kinetic split.:
`L_O_kinetic_eq_sum_of_sector_pieces`.
Hypercharge kinetic aggregate.:
`L_SM_B_kinetic`.
Weak kinetic aggregate.:
`L_SM_W_kinetic`.
Strong kinetic aggregate.:
`L_SM_G_kinetic`.
Gauge–kinetic equality.:
`L_SM_gauge_kinetic_eq_L_O_kinetic`.
SM embedding module.:
`Hqiv/Algebra/SMEmbedding.lean`.
Per-generation minimal coupling.:
`L_SM_fermion_minimal_coupling`.
Three generations from triality.:
`three_generations_from_triality_reps`.
Three-generation equality.:
`L_SM_three_generations_eq_total_source`.
Source-coupling additivity.:
`L_O_source_general_add_J`.
SM–GR unification module.:
`Hqiv/Physics/SM_GR_Unification.lean`.
Generation index map.:
`smGenerationIndex`.
Charged-lepton resonance.:
`Hqiv/Physics/ChargedLeptonResonance.lean`.
Resonance-product factor.:
`resonanceProduct`.
Weak/Higgs scaffold.:
`Hqiv/Physics/WeakHiggsFromOMaxwellScaffold.lean`.
Octonion scalar norm.:
`scalarNormSq`.
Higgs potential.:
`higgsPotential`.
Lock-in vev.:
`lockinVev`.
Effective quartic.:
`lambda_eff`.
SM Higgs kinetic shadow.:
`L_SM_Higgs_kinetic`.
SM Higgs potential shadow.:
`L_SM_Higgs_potential`.
Expanded Higgs potential.:
`L_SM_Higgs_potential_expanded`.

Lock-in vev squared.:
`lockinVev_sq_eq_eta_paper`.
Lock-in Ω_k calibration.:
`omega_k_lockin_calibration`.
Geometric SM mass functional.:
`smMassFromGeometryLabel`.
SM mass-label type.:
`SMMassLabel`.
Two resonance steps.:
`resonance_k_tau_mu`, `resonance_k_mu_e`.
Tau Planck mass.:
`m_tau_Pl`.
SM Yukawa coupling.:
`y_SM`.
Per-flavour Yukawa term.:
`L_SM_Yukawa_flavour`.
All-flavours list.:
`all_SM_flavours`.
All-flavours length.:
`all_SM_flavours_length`.
Yukawa–mass identity.:
`y_SM_times_v_over_sqrt2_eq_mass`.
Yukawa resonance witness.:
`SM_Lagrangian_yukawa_resonance_witness`.
Yukawa sum identity.:
`L_SM_Yukawa_sum_eq_sum`.
SM Lagrangian record.:
`SM_Lagrangian`.
From-HQIV constructor.:
`SM_Lagrangian.fromHQIV`.
SM Lagrangian total.:
`SM_Lagrangian.total`.
SM-from-HQIV discrete-action theorem.:
`SM_Lagrangian_from_HQIV_discrete_action`.
Triality-summed SM current.:
`J_SM_total`.
Total-action split theorem.:
`HQIV_total_action_eq_SM_gauge_fermion_plus_phi_coupling_plus_grav`.
Parameter witness.:
`SM_Lagrangian_parameter_witness`.
Lattice α identity.:
`alpha_eq_3_5`.
Horizon-split γ identity.:
`gamma_eq_2_5`.
GUT coupling identity.:
`alpha_GUT_eq_1_42`.
Proton lock-in scale-witness.:
`ScaleWitness.proton_lockin` in `Hqiv/Physics/ScaleWitness.lean`.

κ_6 factorisation.: `tuftHopfKappa6_eq_matter_fraction_gamma_lapse_concentration` in Hqiv/Physics/HopfShellBeltramiMassBridge.lean.
Local matter fraction on shells.: `tuftMatterFractionAtXi_eq_eta_partial_on_shell`.
Lock-in lapse concentration.: `tuftLapseConcentrationAtXi_lockin_zero`.
κ_6 at horizon.: `tuftHopfKappa6AtXi`, `tuftHopfKappa6`.
TUFT lepton mass at ξ.: `tuftLeptonMassFromVevAtXi_MeV`.

References

[1] Steven Jr Ettinger. Electroweak scale and fermion masses from horizon readouts: First-principles electroweak closure inside the HQIV lean library. Preprint, HQIV-LEAN repository (`papers/lean_to_mass_spectrum/`), 2026. In preparation. To be deposited at Zenodo (HQIV community) on Lean PR build completion; this entry will be updated with the minted DOI on publication.

[2] Steven Jr Ettinger. Horizon-quantized informational vacuum (hqiv): A unified framework from causal horizon monogamy and discrete null-lattice combinatorics. Zenodo preprint, 2026. Version v2.

[3] Steven Jr Ettinger. Kirchhoff's law of thermal emission with built-in UV/IR cutoffs from HQIV's discrete null lattice. Zenodo (HQIV community), 2026. Version v1 (2026-05-27). Tier-1 finite-mode Kirchhoff law; includes `scripts.zip` (birefringence, hockey-stick check, Friedmann recovery). Source: `papers/finite_mode_kirchhoff/`.

[4] Steven Jr Ettinger. Light-cone-derived auxiliary fields supply an explicit derived interpretation of quantum superposition. Zenodo (HQIV community), 2026. Concept DOI; v2 DOI 10.5281/zenodo.19336553 (2026-03-30). Octonionic $\mathbb{R}^8$ carrier, lossless QM bookkeeping, $\alpha = 3/5$ / $\gamma = 2/5$ forced from the lattice.

[5] Steven Jr Ettinger. Octonionic action from the HQIV-LEAN variational layer: Derivation, holonomy alignment, and a tiered uniqueness thesis. Zenodo (HQIV community), 2026. Version v1 (2026-05-27). Manuscript Preprint v3; includes `scripts.zip` (SPARC first-pass table and minimal-seed worked example). Source: `papers/octonionic_action/`.

[6] Steven Jr Ettinger. Review: From discrete null-lattice growth to so(8): Concrete realization and exact lie closure. Zenodo (HQIV community), 2026. Concept DOI; latest version DOI 10.5281/zenodo.20214211 (2026-05-15). Includes `closure.pdf` and `so8_closure_full_appendix.pdf`.

[7] Steven Jr Ettinger. Three-dimensional causal growth and the octonionic gauge sector: Conditional forcing and a uniqueness conjecture. Zenodo (HQIV community), 2026. Version v1 (2026-05-27). Mathematical sequel to `hqiv-so8-paper`; dimensional hinge, conditional forcing, $(\alpha, \gamma) = (3/5, 2/5)$ from unit split + spanning-tree balance. Source: `papers/3d_causal_growth/`.

[8] Steven Jr Ettinger. TUFT/HQIV dynamic mass spectrum: Inner–outer casimir readouts on the hopf carrier. Preprint, HQIV-LEAN repository (`papers/tuft_hqiv_dynamic_mass_spectrum/`), 2026. Dynamic mass-spectrum synthesis tying the TUFT Hopf/Beltrami scalar, vev chart, trapped-Casimir scale, and comparison excited-state readouts.

[9] Ettinger, Steven Jr. and HQIV contributors. Hqiv lean 4 formalization (slimmed public mirror). GitHub (HQIV organization), 2026. Public, slimmed-down Lean 4 development carrying the modules cited from the HQIV paper series. Companion Zenodo manuscript DOI 10.5281/zenodo.18899939.

[10] Michael E. Peskin and Daniel V. Schroeder. *An Introduction to Quantum Field Theory*. Addison-Wesley, 1995.

[11] Matthew D. Schwartz. *Quantum Field Theory and the Standard Model*. Cambridge University Press, 2014.

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